# Notes on Olympiad Geometry

Samuel de Araújo Brandão

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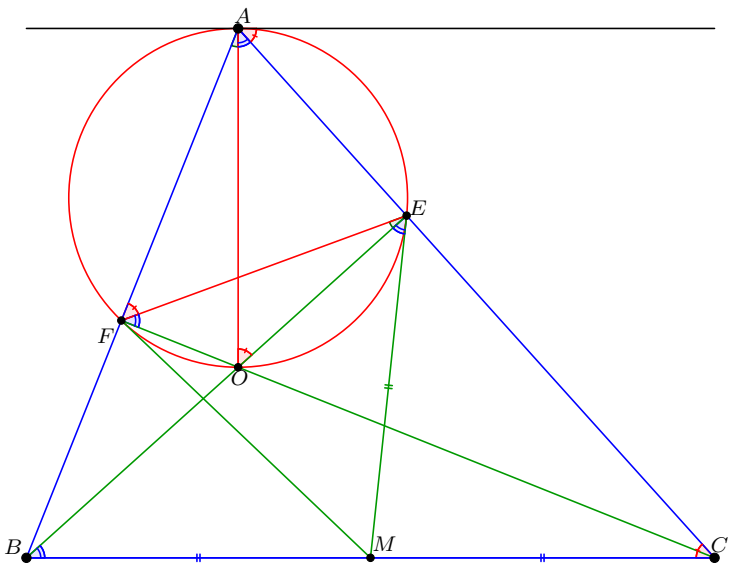
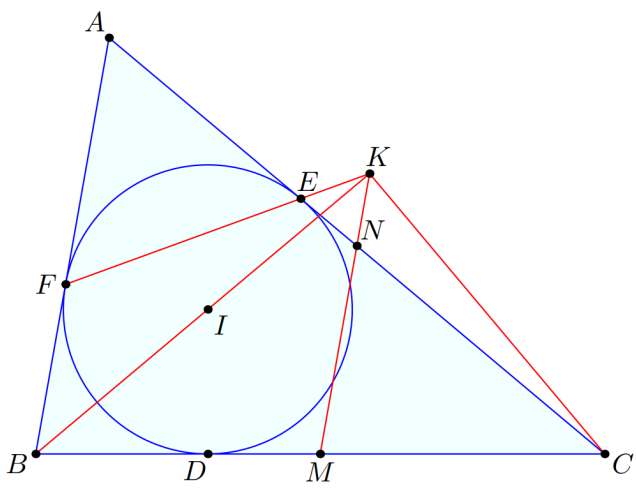
**Lemma 0.1: Three Tangents Lemma**

Let $ABC$ be an acute triangle. Let $BE$ and $CF$ be altitudes of $\triangle ABC$, and denote by $M$ the midpoint of $BC$. Prove that $ME$, $MF$, and the line through $A$ parallel to $BC$ are all tangents to $(AEF)$.




**Lemma 0.2: Iran Lemma**

The incircle of $\triangle ABC$ is tangent to $BC, CA, AB$ at $D, E, F$, respectively. Let $M$ and $N$ be the midpoints of $BC$ and $AC$, respectively. Ray $BI$ meets line $EF$ at $K$. Show that $BK \perp CK$. Then show $K$ lies on line $MN$.




**Solution.**

**Claim** — $BK \perp CK$

*Proof.* Since $F$ and $D$ are reflected over $BI$, the following is true.

$$\measuredangle KDC = \measuredangle BDK = \measuredangle KFB = \measuredangle EFA = \measuredangle AEF = \measuredangle CEK. \qquad \square$$

**Claim** — $K$ lies on line $MN$.

*Proof.* Since $M$ is the circumcenter of $(BKC)$, $\measuredangle CMK = 2\measuredangle CBK = \measuredangle CBA$. Hence, $KM \parallel AB$. However, $MN \parallel AB$. Thus, $N \in KM$. $\square$

### 0.0.1 USAJMO 2011 P5

**Problem Statement**

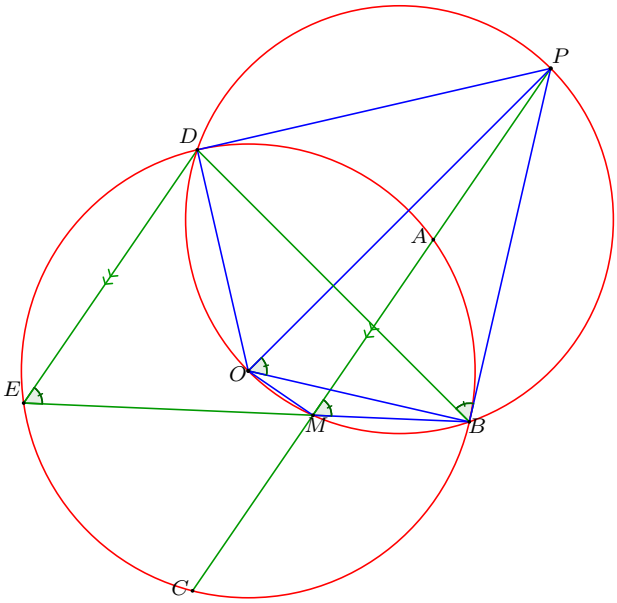
Points $A, B, C, D, E$ lie on a circle $\omega$ and point $P$ lies outside the circle. The given points are such that

(i) lines $PB$ and $PD$ are tangent to $\omega$,

(ii) $P, A, C$ are collinear, and

(iii) $DE \parallel AC$.

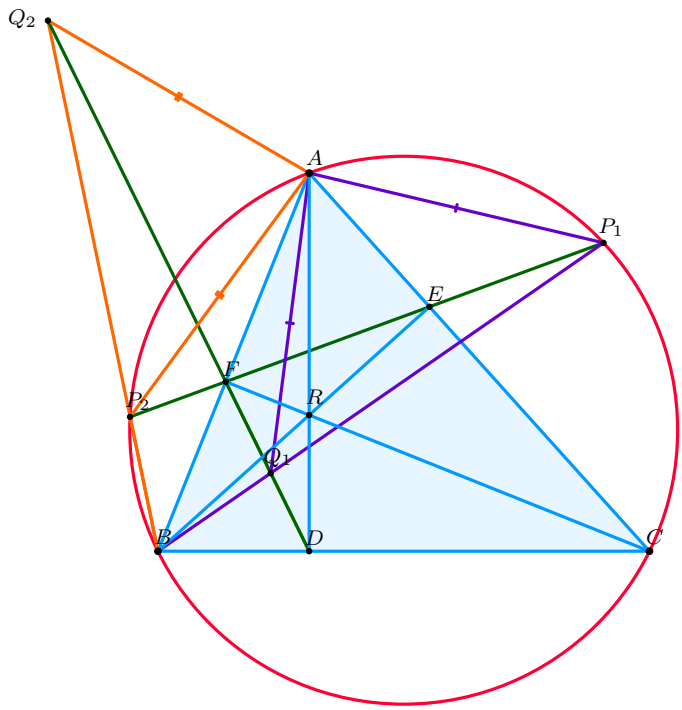
Prove that $BE$ bisects $AC$.



### 0.0.2 Shortlist 2010 G1

**Problem Statement**

Let $ABC$ be an acute triangle with $D, E, F$ the feet of the altitudes lying on $BC, CA, AB$ respectively. One of the intersection points of the line $EF$ and the circumcircle is $P$. The lines $BP$ and $DF$ meet at point $Q$. Prove that $AP = AQ$.




Let $\measuredangle$ denote directed angles mod 180°. The line EF meets the circumcircle at two points. Directed angles allow us to treat both at once, so we fix one of them.

Our goal is to show $\measuredangle PQA = \measuredangle APQ$, i.e., that $\triangle APQ$ is isosceles. We will first prove that $A$, $F$, $P$, $Q$ are concyclic.

Since $\triangle DEF$ is an orthic triangle, $\measuredangle CFA = \measuredangle ADC = 90°$. Therefore, $FACD$ is cyclic, culminating in $\measuredangle ACD = \measuredangle AFD = \measuredangle AFQ$. However, $APCB$ is cyclic, so $\measuredangle APB = \measuredangle ACB = \measuredangle ACD$. Thus, $\measuredangle APQ = \measuredangle AFQ$. Which means that $AFPQ$ is a cyclic quadrilateral.

$FECB$ is a cyclic quadrilateral either, by the same $FACD$'s reason. Therefore, putting together everything we've seen so far:

$$\measuredangle PQA = \measuredangle PFA = \measuredangle EFB = \measuredangle ECB = \measuredangle ACB = \measuredangle APB.$$

Hence, $AP = AQ$.

### 0.0.3 Euler's Lemma

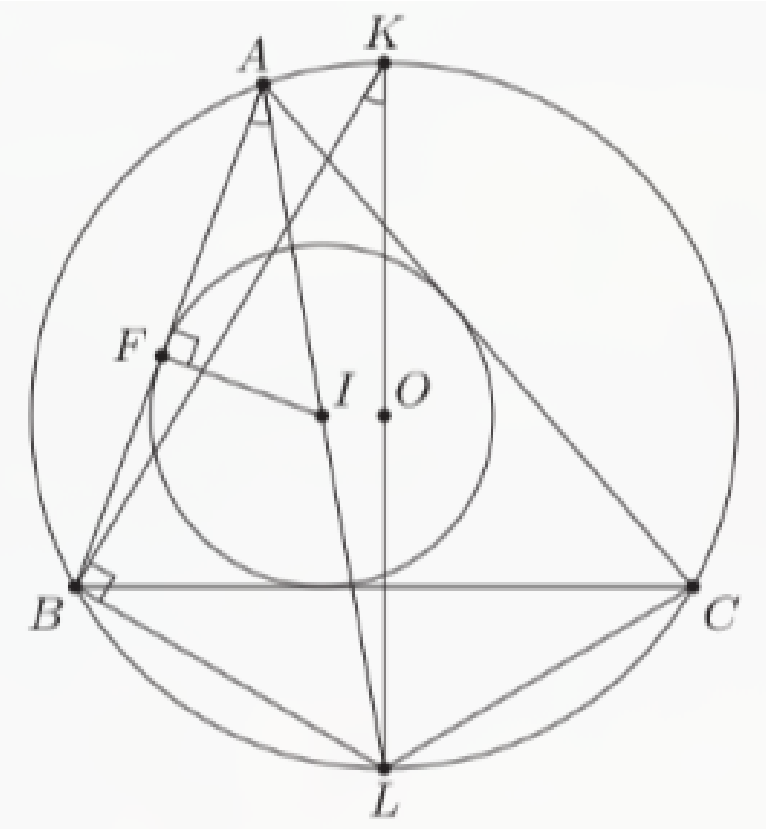



**Figure 2.7C.** Proving Euler's theorem.

*Solution to Lemma 2.22.* Let ray $AI$ meet the circumcircle again at $L$ and let $K$ be the point diametrically opposite $L$. Let $F$ be the foot from $I$ to $\overline{AB}$. Notice that $\angle FAI = \angle BAL = \angle BKL$ and $\angle AFI = \angle KBL = 90^\circ$, so

$$\frac{AI}{r} = \frac{AI}{IF} = \frac{KL}{LB} = \frac{2R}{LI}$$

and hence $AI \cdot IL = 2Rr$. Because $I$ lies inside $\triangle ABC$, we deduce the power of $I$ with respect to $(ABC)$ is $2Rr = R^2 - OI^2$. Consequently, $OI^2 = R(R - 2r)$. $\square$

### 0.0.4 All-Russian MO 2011

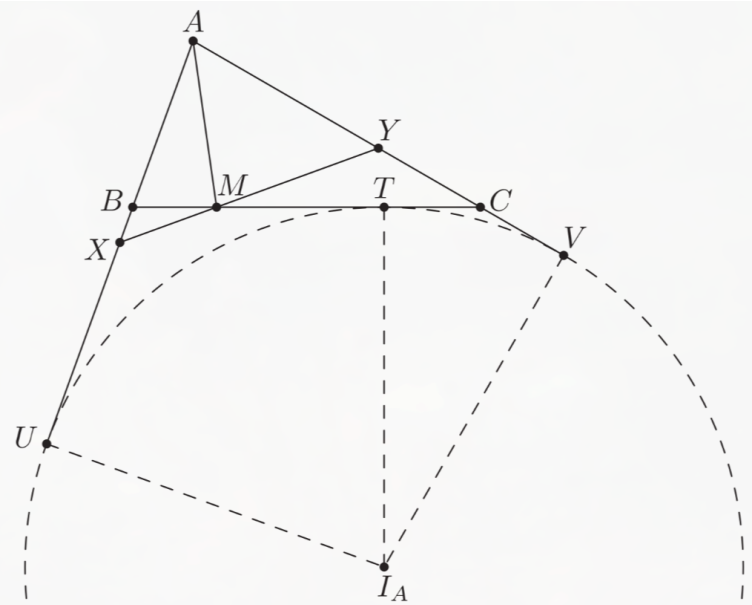



**Figure 2.7E.** Adding an excircle to handle the conditions.

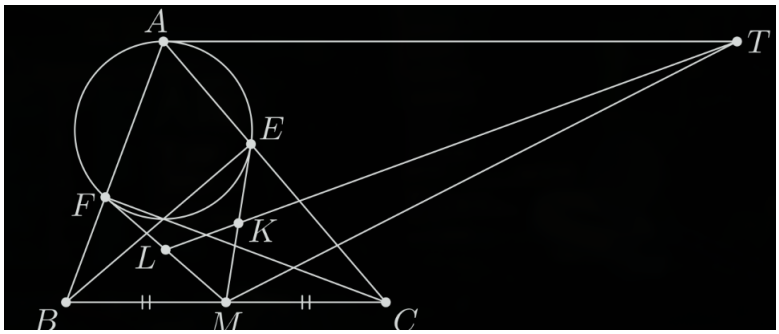
*Solution to Example 2.23.* Let $I_A$ be the center of the $A$-excircle, tangent to $\overline{BC}$ at $T$, and to the extensions of $\overline{AB}$ and $\overline{AC}$ at $U$ and $V$. We see that $AU = AV = s = 2$. Then $\overline{XY}$ is the radical axis of the $A$-excircle and the circle of radius zero at $A$. Therefore $AM = MT$.

Assume without loss of generality that $T$ lies on $\overline{MC}$, as opposed to $\overline{MB}$. Then $AB + BM + MA = AB + BM + MT = AB + BT = AB + BU = AU = 2$ as desired. $\square$

### 0.0.5 Iran TST 2011/1 02/13/2026

**Problem Statement**

In acute triangle $ABC$, $\angle B > \angle C$. Let $M$ be the midpoint of $\overline{BC}$ and let $E$ and $F$ be the feet of the altitudes from $B$ and $C$, respectively. Let $K$ and $L$ be the midpoints of $\overline{ME}$ and $\overline{MF}$, respectively, and let $T$ lie on line $KL$ such that $\overline{TA} \parallel \overline{BC}$. Prove that $TA = TM$.




Let $\Gamma_1 = (AFE)$ and $\Gamma_2$ be the circumference of radius 0 centered at $M$. In order to prove $TA = TM$, it is enough to demonstrate that $\text{Pow}_{\Gamma_1}(T) = \text{Pow}_{\Gamma_2}(T)$.

**Claim** — $\overline{KL}$ is the radical axis of $\Gamma_1$ and $\Gamma_2$.

*Proof.* By the Three Tangents Lemma, $\overline{ME}$ and $\overline{MF}$ are both tangents to $\Gamma_1$. Since, $MK = KE$,

$$\text{Pow}_{\Gamma_1}(K) = \text{Pow}_{\Gamma_2}(K).$$

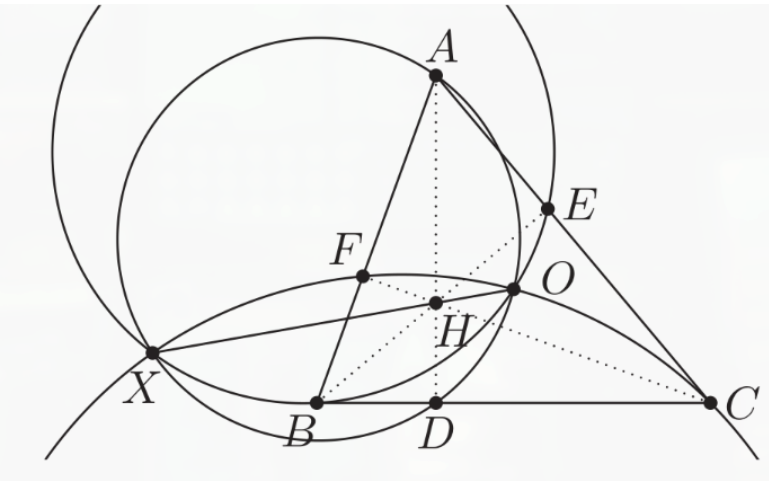
Hence, $K$ lies inside the radical axis of $\Gamma_1$ and $\Gamma_2$. However, we can use an analogous argument to prove that $L$ does either. Hence $\overline{KL}$ is the radical axis of $\Gamma_1$ and $\Gamma_2$. $\square$

Since $K$, $L$ and $M$ are collinear, $M$ lies inside this radical axis too. Hence, $TA = TM$

### 0.0.6 02/13/26

**Problem Statement**

Let $\overline{AD}$, $\overline{BE}$, $\overline{CF}$ be the altitudes of a scalene triangle $ABC$ with circumcenter $O$. Prove that $(AOD)$, $(BOE)$, and $(COF)$ intersect at point $X$ other than $O$.




Let $H$ be the orthocenter $\triangle ABC$. Hence,

$$HA \cdot HD = HB \cdot BE = HC \cdot HF,$$

is true by considering the circumcenters of diameters $AB$, $BC$ and $CA$. Which means that both $O$ and $H$ have the same power of point of all the circumferences $(AOD)$, $(BOE)$, and $(COF)$.

Hence, those three circumferences have the same radical axis, resulting in another point $X$ where they three intersect.